Cardiovascular System (CARD) · NPLEX-P1-CARD-18

CV-18-G Congenital Heart Conditions

Work through each activity on blank paper with the notes chapter closed, then open the answer key and mark what you missed. Do the two brain dumps for a competency on different days rather than back to back.

Work each activity without your notes first. Rate your confidence before you check. Then read the notes, write down what you missed, and set a date to come back to it.

Competency tracker

Competency	BD1 done	BD2 done	Confidence 1 to 4	Next review date
NPLEX-P1-CARD-18-07-C01				
NPLEX-P1-CARD-18-07-C02				
NPLEX-P1-CARD-18-07-C03				
NPLEX-P1-CARD-18-07-C04				
NPLEX-P1-CARD-18-07-C05				
NPLEX-P1-CARD-18-07-C06				
NPLEX-P1-CARD-18-07-C07				
NPLEX-P1-CARD-18-07-C08				

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-18-07-C01-BD1

Draw the three left-to-right shunts and label the pressures

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-G Congenital Heart Conditions
Competency: NPLEX-P1-CARD-18-07-C01 Activity: NPLEX-P1-CARD-18-07-C01-BD1 Date: _____

Without looking at your notes, draw a four-chamber heart with the aorta and pulmonary trunk, then add a ventricular septal defect, an atrial septal defect, and a patent ductus arteriosus to the same drawing. Write the normal pressure in each ventricle after birth and mark the direction blood crosses each defect with an arrow. Beside each arrow write what makes blood move that way, naming pressure for one defect and chamber compliance for another. Then write what Qp:Qs means, what value it takes with a net left-to-right shunt, and why systemic arterial saturation stays normal in all three. Finish by writing which of the three defects transmit systemic pressure to the pulmonary arteries and which adds only extra volume, and explain the difference from where each defect sits relative to the tricuspid valve.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-18-07-C01-BD2

Explain every Eisenmenger finding from the reversal

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-G Congenital Heart Conditions
Competency: NPLEX-P1-CARD-18-07-C01 Activity: NPLEX-P1-CARD-18-07-C01-BD2 Date: _____

A 29-year-old woman has an uncorrected ventricular septal defect and is now cyanotic. Write what has happened to her pulmonary arterioles, naming the structural changes in the order they develop and the two endothelial mediators that fall and the one that rises. Then take each of these findings and write in one or two sentences why it is present: cyanosis that appears only now, cyanosis that barely improves on supplemental oxygen, a raised red cell mass, clubbed fingers, a loud P2, and a murmur that is softer than it was ten years ago. Write which chamber has hypertrophied and what raised its pressure. Finish by writing why a child with an atrial septal defect of the same size would be expected to reach this point later, and whether the arteriolar changes at this stage can reverse.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Retrieval exercise NPLEX-P1-CARD-18-07-C01-RT01

Rebuild the path from shunt to reversal

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-G Congenital Heart Conditions
Competency: NPLEX-P1-CARD-18-07-C01 Activity: NPLEX-P1-CARD-18-07-C01-RT01 Date: _____

Rebuild the six-step sequence that runs from a left-to-right shunt to late cyanosis, one step per line and in order. Beside the step where resistance rises, list the three arteriolar changes the notes name and write which one marks irreversible disease. Beside the step where the shunt reverses, write the pressure comparison that must be true for reversal to happen. Underneath the sequence, draw the shunt arrow at three points in time: early, when the shunt is bidirectional, and after reversal, and write the patient's systemic arterial saturation at each point.

Rebuild from memory: From left-to-right shunt to Eisenmenger syndrome

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Prediction exercise NPLEX-P1-CARD-18-07-C01-PR01

Predict what happens as pulmonary vascular resistance falls after birth

A term infant is born with a large ventricular septal defect. Over the first eight weeks pulmonary vascular resistance falls from its high fetal value toward the normal postnatal value. Predict each item below, write the direction of the change, and give the pressure, resistance or volume reason in one line.

What changes	Up, down, or no change	Why, in one or two lines
What happens to the volume of blood crossing the defect from left to right		
What happens to Qp:Qs		
What happens to pulmonary blood flow and to left atrial and left ventricular volume		
What happens to the loudness of the murmur in a defect that is large and nonrestrictive		
What the infant looks like clinically by the second month, listing the findings of heart failure the notes give		
What happens to systemic arterial oxygen saturation during these eight weeks, and why		

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-18-07-C02-BD1

Build the ventricular septal defect from the septum outward

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-G Congenital Heart Conditions
Competency: NPLEX-P1-CARD-18-07-C02 Activity: NPLEX-P1-CARD-18-07-C02-BD1 Date: _____

Draw the interventricular septum and label its muscular part and its membranous part, marking where the membranous part sits relative to the aortic valve. Write the three structures whose fusion forms the membranous septum and write why that part is the one most often left incomplete. Name the two main locations of ventricular septal defects and say which is more common. Then write the murmur in full: its timing across the cardiac cycle, its quality, where it is loudest by intercostal space, what else may be felt there, and what sustained handgrip does to it and why. Finish by writing which chambers carry the extra volume and enlarge, and explain from where in the cycle the shunted blood leaves the left ventricle why the right ventricle is spared until pulmonary pressure rises. List the associations the notes give.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-18-07-C02-BD2

Two infants, two murmurs, two courses

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-G Congenital Heart Conditions
Competency: NPLEX-P1-CARD-18-07-C02 Activity: NPLEX-P1-CARD-18-07-C02-BD2 Date: _____

Two infants each have a ventricular septal defect. Infant A has a loud harsh holosystolic murmur with a thrill and is feeding and growing well. Infant B has a soft short systolic murmur, an apical mid-diastolic rumble, a loud P2, tachypnea and sweating with feeds, and poor weight gain. For each infant write which defect is the larger one and justify it from the pressure gradient across the defect and from right ventricular pressure. Write what sets the size of the shunt in each. Then write the expected course of each over childhood, including which is likely to close on its own and by what mechanism, and which is at risk of irreversible pulmonary vascular disease and roughly how soon. Finish with two complications of a defect sitting just below the aortic valve, and write the mechanism of each.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-18-07-C03-BD1

Build the atrial septum, then explain the fixed split S2

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-G Congenital Heart Conditions
Competency: NPLEX-P1-CARD-18-07-C03 Activity: NPLEX-P1-CARD-18-07-C03-BD1 Date: _____

Draw the developing atrial septum from memory and label the septum primum, the septum secundum, the ostium primum, the ostium secundum, and the foramen ovale, marking which structure acts as the flap valve. Write what holds the flap open in the fetus and what presses it shut after birth. Then write why an atrial septal defect shunts left to right even though left and right atrial pressures differ by only a few mmHg, naming the property of the right ventricle that is responsible. Write which chambers and vessels receive the extra volume and enlarge. Finish by explaining the two auscultatory findings: say why flow across the defect itself is silent, where the systolic murmur actually comes from and where it is heard, and then explain step by step why the second heart sound is widely split and stays split in both inspiration and expiration.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-18-07-C03-BD2

A stroke at 34 with no murmur

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-G Congenital Heart Conditions
Competency: NPLEX-P1-CARD-18-07-C03 Activity: NPLEX-P1-CARD-18-07-C03-BD2 Date: _____

A 34-year-old man with no cardiac history has a stroke two days after a long flight, and a deep vein thrombosis is found in his leg. Write the cardiac finding that would explain the stroke and write why it caused neither a murmur nor any chamber enlargement in the years before this. Write the pressure relationship that normally keeps it closed and list the everyday events the notes name that reverse that relationship for a moment. Then write why the same event can send a clot from the leg into the systemic arteries, and name that process. Separately, write the three adult presentations of an atrial septal defect that was never found in childhood, and for the rhythm problem write which chamber change produces it. Finish by writing which atrial lesion involves missing septal tissue and which involves tissue that simply never sealed, and which of the two is associated with a cleft mitral leaflet.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Comparison table NPLEX-P1-CARD-18-07-C03-CP01

The three lesions of the atrial septum

Fill in every cell from memory, then open the notes chapter and correct anything you missed.

Feature	Ostium secundum ASD	Ostium primum ASD	Patent foramen ovale
What went wrong in development			
Is septal tissue missing, or is complete tissue unsealed			
How common it is			
Genetic or exposure associations			
Associated valve abnormality, if any			
Direction of the axis on the ECG, and any conduction finding			
Whether there is a resting shunt large enough to overload the right heart			

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-18-07-C04-BD1

The ductus arteriosus from arch to ligament

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-G Congenital Heart Conditions
Competency: NPLEX-P1-CARD-18-07-C04 Activity: NPLEX-P1-CARD-18-07-C04-BD1 Date: _____

Write which aortic arch the ductus arteriosus comes from, then draw the aortic arch and pulmonary trunk and show exactly where the ductus attaches at each end, naming the branch of the arch it lies just distal to. Write what the ductus does in fetal circulation and why so little right ventricular output goes to the lungs. Name the two signals that keep its smooth muscle relaxed before birth and write where the prostaglandin comes from and why the fetus does not clear it. Then rebuild the five-step sequence of closure after birth in order, and write what the closed ductus becomes. Finish by naming the risk factors for a ductus that stays open and writing the mechanism for each, and name the nerve that hooks under the ligamentum arteriosum.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-18-07-C04-BD2

Pulses, pressures and blue toes

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-G Congenital Heart Conditions
Competency: NPLEX-P1-CARD-18-07-C04 Activity: NPLEX-P1-CARD-18-07-C04-BD2 Date: _____

A term infant has a continuous machine-like murmur below the left clavicle, a wide pulse pressure and bounding pulses. Write the pressure comparison between the aorta and the pulmonary artery in systole and in diastole, and use it to explain why the murmur never stops and why it peaks where it does. Then explain the pulse findings in two parts: why diastolic pressure falls, and why systolic pressure rises. Write which chambers dilate and why. Next, write why a premature infant with the same ductus may have a murmur only in systole. Finally, this patient reaches adulthood uncorrected and develops Eisenmenger physiology. Write which limbs become cyanotic, which stay pink, and justify the pattern from the attachment point of the ductus on the arch, then name that finding.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Retrieval exercise NPLEX-P1-CARD-18-07-C04-RT01

Rebuild ductal closure after birth

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-G Congenital Heart Conditions
Competency: NPLEX-P1-CARD-18-07-C04 Activity: NPLEX-P1-CARD-18-07-C04-RT01 Date: _____

Rebuild the five-step sequence of ductal closure after birth, one step per line and in order. Beside the oxygen step write what rising arterial PO2 does to ductal smooth muscle, and beside the prostaglandin step write which prostaglandin falls and what two changes at birth lower it. Beside the last step write the name of the remaining structure and how long anatomic closure takes. Underneath, write what happens to the ductus when cyclooxygenase is inhibited and what happens when prostaglandin E1 is given, and then write which of these two findings explains why a premature infant is at the highest risk.

Rebuild from memory: Closure of the ductus arteriosus after birth

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-18-07-C05-BD1

Build tetralogy of Fallot from one developmental error

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-G Congenital Heart Conditions
Competency: NPLEX-P1-CARD-18-07-C05 Activity: NPLEX-P1-CARD-18-07-C05-BD1 Date: _____

Write the single developmental error that produces tetralogy of Fallot, naming the septum involved and the direction it is displaced. Then list the four features and write beside each one how that displacement produces it, making clear which feature is a consequence of the other three rather than a primary defect. Draw the right ventricular outflow tract, the septal defect and the aortic root in one sketch and label all four features on the drawing. Write which cell population builds the outflow septum and name the deletion syndrome that follows from its failure, with the two other organs that syndrome affects and the two other cardiac lesions it is linked to. Finish with the other associations the notes give.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-18-07-C05-BD2

Why one child with tetralogy is blue and another is not

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-G Congenital Heart Conditions
Competency: NPLEX-P1-CARD-18-07-C05 Activity: NPLEX-P1-CARD-18-07-C05-BD2 Date: _____

Two children have tetralogy of Fallot. One is deeply cyanotic from the first weeks and the other has almost normal saturation at age four. Write the one variable that differs between them and explain, using the two exits available to the right ventricle and the resistance of each, how that variable decides the saturation. Write why the ventricular septal defect itself produces no murmur, where the murmur in this lesion actually comes from, and why S2 is single. Write why pulmonary vascular disease and Eisenmenger reversal do not develop here. Then the second child has a hot bath, cries hard, and becomes deeply cyanotic while the murmur fades. Write the five-step loop that produced the spell, write why the murmur softened, and write what squatting does to systemic vascular resistance and to the share of right ventricular output reaching the lungs. Finish by naming two findings of chronic hypoxemia and two consequences of right-to-left flow that the notes give.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-18-07-C06-BD1

Bicuspid aortic valve and its four complications

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-G Congenital Heart Conditions
Competency: NPLEX-P1-CARD-18-07-C06 Activity: NPLEX-P1-CARD-18-07-C06-BD1 Date: _____

Write how common a bicuspid aortic valve is, which sex it favors, and which two cusps most often fail to separate, then draw the valve from above showing two unequal leaflets and label the ridge that marks the fused commissure. Name the gene the notes give for some familial cases and the two conditions that cluster with it. Then write the four complications of this valve and for each one write its mechanism in one or two sentences, making clear which complication dominates in younger patients and which in older ones. Write the sequence of tissue changes that leads to calcific stenosis, and write roughly how much earlier significant stenosis appears than with a three-cusp valve. Finish by writing the one auscultatory clue the notes give for a mobile bicuspid valve and why it disappears later.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-18-07-C06-BD2

Why the aorta is part of the same disease

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-G Congenital Heart Conditions
Competency: NPLEX-P1-CARD-18-07-C06 Activity: NPLEX-P1-CARD-18-07-C06-BD2 Date: _____

A 38-year-old man with a known bicuspid aortic valve is found to have a dilated ascending aorta even though the valve is working well. Write the two reasons the notes give for the aortic wall changes, saying which one is intrinsic to the wall and which comes from flow, and name the two histologic changes in the media. Write why the finding can appear while the valve still functions normally, and name the two complications the dilated aorta can lead to. Then write which two coexisting conditions raise that risk further. Separately, explain why this patient has a higher risk of infective endocarditis than someone with a three-cusp valve, naming what forms on the leaflet surface first. Finish by writing why a 22-year-old with the same valve is more likely to present with a regurgitant murmur while a 62-year-old presents with a stenotic one, giving a mechanism for each.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-18-07-C07-BD1

Coarctation: where it sits and what it does to the pulses

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-G Congenital Heart Conditions
Competency: NPLEX-P1-CARD-18-07-C07 Activity: NPLEX-P1-CARD-18-07-C07-BD1 Date: _____

Draw the aortic arch with its three branches and mark exactly where a coarctation sits, naming the branch it lies just distal to and the structure its narrowing is tied to. Write the leading explanation for why the narrowing forms there. Name the sex it favors, the chromosomal syndrome it accompanies, and the valve lesion present in most patients. Then write what a fixed high resistance in series does to pressure upstream and to pressure and flow downstream, and from that write the full examination picture: arm blood pressure, leg blood pressure and how the two normally compare, the femoral pulse in strength and in timing relative to the radial pulse, and what the legs feel like with exercise. Write which hormonal system is activated and why. Finish by listing the complications of long-standing hypertension proximal to the narrowing that the notes give, and name which chamber hypertrophies.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-18-07-C07-BD2

The newborn who collapses and the teenager who does not

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-G Congenital Heart Conditions
Competency: NPLEX-P1-CARD-18-07-C07 Activity: NPLEX-P1-CARD-18-07-C07-BD2 Date: _____

A newborn with coarctation is well on day one and in shock with absent femoral pulses on day six. A 15-year-old with coarctation is found on a sports physical with hypertension and no symptoms. For each patient write how severe the narrowing is, what the lower body was being perfused by before the problem appeared, and what event or passage of time produced the presentation. For the newborn write why lower body cyanosis can be present, what happens to the kidneys when the ductus closes, and why acidosis follows. For the teenager write what has developed over years that prevents collapse, then rebuild the five-step collateral route in order from the vessel above the narrowing to the aorta below it, naming the one segment in which flow runs backwards. Finish by writing the three chest film findings the notes give, why the first two ribs are spared from notching, why notching is rarely seen in a young child, and where the murmur of collateral and coarctation flow is heard.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Retrieval exercise NPLEX-P1-CARD-18-07-C07-RT01

Rebuild the collateral route around a coarctation

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-G Congenital Heart Conditions
Competency: NPLEX-P1-CARD-18-07-C07 Activity: NPLEX-P1-CARD-18-07-C07-RT01 Date: _____

Draw the aortic arch, the coarctation, and the chest wall, then rebuild the five-step collateral route in order from the vessel above the narrowing to the descending aorta below it, labelling each vessel on your drawing. Mark the one segment in which flow runs in the reverse of its normal direction. Beside the drawing write which vessels enlarge over years, which bony structures they erode and where on those structures, and the radiographic name of that finding. Then write which two ribs are spared and give the arterial reason, and write why this finding is unusual in a young child.

Rebuild from memory: Collateral flow around a coarctation

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Clinical application NPLEX-P1-CARD-18-07-C07-CA01

Hypertension in a 16-year-old with weak femoral pulses

A 16-year-old girl is seen for a sports physical. She has no symptoms other than aching calves after long runs. She is 147 cm tall with a webbed neck and a broad chest. Temperature is 36.8 degrees C (98.2 degrees F), pulse is 76/min and regular, and respirations are 14/min. Blood pressure is 158/92 mm Hg in the right arm and 104/64 mm Hg in the right leg. The radial pulses are strong and the femoral pulses are weak and arrive slightly after the radial pulse. A systolic murmur is heard over the back between the scapulae, and a separate early systolic click and soft systolic murmur are heard at the right upper sternal border. The chest film shows notching along the inferior margins of the fourth through eighth ribs and an indented aortic contour with dilation above and below it.

1. Recognize. Name the aortic lesion, state where in the aorta it sits, and name the second cardiac lesion the click and upper sternal murmur point to and the chromosomal syndrome her growth and physical features suggest.

2. Explain. Explain from flow, pressure and resistance why her arm pressure is high while her leg pressure is low, and explain why the femoral pulse is both weak and delayed.

3. Differentiate. Write how this pressure and pulse pattern differs from the two other causes of leg desaturation or weak lower body perfusion in this chapter, a patent ductus arteriosus with Eisenmenger physiology and a severe neonatal coarctation, naming the findings that separate them.

4. Investigate. Write what the rib notching and the indented aortic contour tell you, which vessels produce the notching, and which measurements on examination you would record to document the gradient.

5. Decide. State which hormonal system is adding to her hypertension and name the three complications of long-standing hypertension proximal to the narrowing that she is at risk of.

6. Justify. Justify the renal part of your answer from perfusion pressure at the kidney, and justify the dissection risk from both the pressure load and the aortic wall changes that accompany her valve lesion.

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-18-07-C08-BD1

Draw the parallel circulations of transposition

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-G Congenital Heart Conditions
Competency: NPLEX-P1-CARD-18-07-C08 Activity: NPLEX-P1-CARD-18-07-C08-BD1 Date: _____

Write what the conotruncal ridges normally do as they grow, including how far the aorticopulmonary septum twists and which ventricle each great vessel ends up connected to. Then write what goes wrong in transposition and which vessel arises from which ventricle, with the usual position of the aorta. Draw both circuits as two separate loops and label, for each loop, the chambers in order, the vessel leaving the heart, and whether the blood in it is saturated or desaturated. Write the four-step path through each loop and then write the fifth line: what must be present for the infant to survive and the three communications that can provide it. Name the maternal risk factor and the sex this lesion favors. Finish by writing why the circulations are described as parallel rather than in series, and what that means for the tissues.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-18-07-C08-BD2

Why oxygen barely helps and why day three is worse

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-G Congenital Heart Conditions
Competency: NPLEX-P1-CARD-18-07-C08 Activity: NPLEX-P1-CARD-18-07-C08-BD2 Date: _____

A newborn with transposition and an intact ventricular septum is mildly cyanotic at birth and deeply cyanotic on day three, and 100% inspired oxygen raises his saturation only slightly. Write why the fetus had no difficulty before delivery. Write which two communications he was depending on after birth and which one is now closing. Then explain why supplemental oxygen barely changes systemic saturation, naming which loop it oxygenates and where that blood goes. Write why mixing must work in both directions rather than one. Next, compare him with an infant who has transposition plus a large ventricular septal defect: write which one is less cyanotic, which one develops heart failure earlier, and give the reason for each. Finish by writing which congenital lesion is the most common cause of cyanosis on the first day of life and which is the most common cyanotic lesion overall, and write what makes the upper mediastinum look narrow on the chest film and what that appearance is called.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Comparison table NPLEX-P1-CARD-18-07-C08-CP01

Tetralogy of Fallot against transposition of the great vessels

Fill in every cell from memory, then open the notes chapter and correct anything you missed.

Feature	Tetralogy of Fallot	Transposition of the great vessels
Developmental error in the outflow septum		
Genetic or maternal associations		
Arrangement of the circulations and the route of desaturated blood into the aorta		
When cyanosis appears and what sets its severity		
Pulmonary blood flow		
Change in saturation with supplemental oxygen		
Chest film appearance		
Whether survival depends on a patent ductus or an atrial communication		

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Board-style question NPLEX-P1-CARD-18-07-C02-Q01

The loud murmur and the quiet one

A 3-week-old infant is brought to the physician for a well child visit. He feeds well and has gained weight appropriately. Temperature is 36.9 degrees C (98.4 degrees F), pulse is 140/min, and respirations are 38/min. A harsh holosystolic murmur with a palpable thrill is heard at the left lower sternal border, and the second heart sound is normal. A second infant of the same age has a soft short systolic murmur at the same location, a loud pulmonic component of the second heart sound, an apical mid-diastolic rumble, tachypnea with feeds, and poor weight gain. Which of the following best explains why the first infant's murmur is louder?

A. A restrictive defect maintains a large systolic pressure gradient between the ventricles

B. A larger defect allows a greater volume of blood to cross the septum each beat

C. Right ventricular pressure has risen to approach left ventricular pressure

D. Pulmonary vascular resistance has already fallen below systemic vascular resistance

Why did you pick it?

Board-style question NPLEX-P1-CARD-18-07-C05-Q02

Squatting during a spell

A 2-year-old boy with tetralogy of Fallot becomes deeply cyanotic and hyperpneic after a prolonged crying episode, and the systolic murmur at his left upper sternal border becomes much softer. His parents place him in a knee-chest position and the cyanosis improves. Which of the following best explains the improvement?

A. Pulmonary vascular resistance falls, drawing more blood through the lungs

B. Venous return from the legs increases, raising right ventricular preload

C. Systemic vascular resistance rises, so a greater share of right ventricular output enters the pulmonary outflow tract

D. The infundibular obstruction relaxes, widening the right ventricular outflow tract

Why did you pick it?

Board-style question NPLEX-P1-CARD-18-07-C04-Q03

Blue toes and pink fingers

A 31-year-old woman with an uncorrected patent ductus arteriosus has become cyanotic over the past several years. Oxygen saturation is 97% on the right hand and 81% on the right foot. The toes are clubbed and the fingers are not. Which of the following best explains the difference in saturation between her hand and her foot?

A. Oxygen extraction is greater in the legs than in the arms because of their larger muscle mass

B. The left ventricle ejects preferentially into the brachiocephalic vessels during early systole

C. Pulmonary venous blood reaches the arch vessels before it reaches the descending aorta

D. Desaturated pulmonary arterial blood enters the aorta distal to the left subclavian artery

Why did you pick it?

Board-style question NPLEX-P1-CARD-18-07-C03-Q04

A second sound that does not move

A 28-year-old woman is evaluated for 6 months of exertional fatigue. A systolic ejection murmur is heard at the left upper sternal border, and the second heart sound is widely split, with the splitting unchanged between inspiration and expiration. The electrocardiogram shows right axis deviation and an incomplete right bundle branch block pattern. Oxygen saturation on room air is 98%. Which of the following best explains the unchanging split of the second heart sound?

A. Pulmonic valve closure is delayed by elevated pulmonary arterial pressure throughout the cycle

B. Right ventricular stroke volume stays nearly constant because inspiratory venous return is offset by reduced shunting

C. Aortic valve closure occurs early because left ventricular stroke volume is reduced

D. Conduction delay in the right bundle branch separates the two components by a fixed interval

Why did you pick it?